

CNC Coolant System — Troubleshooting SOP

Failure signature

A coolant-flow problem usually causes temperature to rise while vibration remains near baseline. Pressure may fall as the pump or filter becomes restricted.

Diagnostic steps

Check coolant level, filter differential pressure, pump current, pressure at the supply manifold, and return flow. Do not replace a spindle bearing solely because of a thermal alert when vibration is stable.

Corrective action

If pump pressure is below the operating range, inspect the filter and pump. Replace COOLANT-PUMP-4KW when pump degradation is confirmed.

Maintenance interval

Inspect coolant filtration weekly and verify pump performance during the scheduled preventive maintenance window.